

Finance

Semantic Model Build — Phase 1 Complete

SECTION 01

Executive Summary

This report documents the successful completion of Phase 1 of the AI-powered reporting automation pipeline for **Finance**. The semantic model has been fully constructed, validated, and is ready for export.

100	30	10	4
Rows processed	Measures built	Column mappings	Issues resolved
1 raw file(s)	from protocol document	7 files joined	4 user • 0 auto

Status	Model is production-ready. All 30 measures have been generated and DAX-validated. 4 data quality decisions were logged with full audit trail.
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SECTION 02

Input Files

Raw data files

File	Rows	Columns	Size (MB)	PK candidates
Retail_FACT_Orders.xlsx	100	31	0.03	OrderLineID

Mapping and dimension files

File	Rows	Columns	Size (MB)	PK candidates
Retail_DIM_Channels.xlsx	8	5	0.01	ChannelID, ChannelName
Retail_DIM_Customers.xlsx	2,000	10	0.13	CustomerID, CustomerName, Email
Retail_DIM_Date.xlsx	1,461	10	0.08	DateKey, Date
Retail_DIM_Products.xlsx	500	10	0.05	ProductID, ProductName

Retail_DIM_Promotions.xlsx	9	8	0.01	PromoID, PromoName
Retail_DIM_ReturnReasons.xlsx	9	4	0.01	ReasonCode, ReasonDesc

Protocol document

Filename	Measures extracted	Clarifications required
Retail_protocol_Document.xlsx	30	16

SECTION 03

Automapping Results

The automapping engine ran a three-pass analysis across all uploaded files. 5 columns matched exactly, 1 by fuzzy matching, 4 by AI semantic inference. 0 column(s) remained unmapped.

Source	Maps to	Conf.	Method	Status
retail_fact_orders.DateKey	retail_dim_date.DateKey	100%	EXACT	Confirmed
retail_fact_orders.CustomerID	retail_dim_customers.CustomerID	100%	EXACT	Confirmed
retail_fact_orders.ProductID	retail_dim_products.ProductID	100%	EXACT	Confirmed
retail_fact_orders.ChannelID	retail_dim_channels.ChannelID	100%	EXACT	Confirmed
retail_fact_orders.PromoID	retail_dim_promotions.PromoID	100%	EXACT	Confirmed
retail_fact_orders.ReturnReasonCode	retail_dim_returnreasons.ReasonCode	90%	FUZZY	Confirmed
retail_fact_orders.OrderDate	retail_dim_date.Date	98%	AI_INFERRED	Confirmed
retail_fact_orders.ShipDate	retail_dim_date.Date	98%	AI_INFERRED	Confirmed
retail_fact_orders.DeliveryDate	retail_dim_date.Date	98%	AI_INFERRED	Confirmed
retail_fact_orders.ReturnReasonCode	retail_dim_returnreasons.ReasonCode	99%	AI_INFERRED	Confirmed

SECTION 04

Generated Measures

30 measures were extracted from the protocol document and converted to DAX formulas. All formulas have been validated for syntactic correctness.

Measure	Aggregation	Format	KPI	Conf.	DAX (summary)
Total Gross Revenue	SUM	\$\$, ##0	✓	100%	Total Gross Revenue = SUM('retail_fact_orders'[GrossRevenue])...
Total Net Revenue	SUM	\$\$, ##0	✓	100%	Total Net Revenue = SUM('retail_fact_orders'[NetRevenue])
Total Orders	COUNTD	#, ##0	✓	100%	Total Orders = DISTINCTCOUNT('retail_fact_orders'[OrderID])
Total Units Sold	SUM	#, ##0	—	100%	Total Units Sold = SUM('retail_fact_orders'[Quantity])
Average Order Value (AOV)	RATIO	\$\$, ##0	✓	100%	Average Order Value (AOV) = DIVIDE(SUM('retail_fact_orders'[...)
Gross Profit	CUSTOM	\$\$, ##0	✓	100%	Gross Profit = SUM('retail_fact_orders'[NetRevenue]) - SUM('...
Gross Margin %	RATIO	0.0%	✓	100%	[Gross Margin %] = DIVIDE([Gross Profit], [Total Net Revenue]...
Discount Rate	RATIO	0.0%	—	100%	Discount Rate = DIVIDE(SUM('retail_fact_orders'[DiscountAmou...
Return Rate	RATIO	0.0%	✓	90%	Return Rate = DIVIDE(CALCULATE(COUNTROWS(r etail_fact_orders)...
Customer Acquisition Count	COUNTD	#, ##0	✓	100%	Customer Acquisition Count = CALCULATE(DISTINCTCOUNT('retail...
Repeat Purchase Rate	RATIO	0.0%	✓	80%	Repeat Purchase Rate = DIVIDE(CALCULATE(DISTINCTCOUNT('retai...
Revenue per Customer	RATIO	\$\$, ##0	—	100%	Revenue per Customer = DIVIDE(SUM('retail_fact_orders'[NetRe...
Category Revenue Share	RATIO	0.0%	—	90%	Category Revenue Share = DIVIDE(SUM('retail_fact_orders'[Net...
Top Product by Revenue	CUSTOM	\$\$, ##0	—	90%	Top Product by Revenue = VAR ProductRevenue = SUMMARIZ...
Channel Revenue Contribution	RATIO	0.0%	—	90%	Channel Revenue Contribution = DIVIDE(SUM('retail_fact_order...

Promotion Effectiveness	RATIO	0.0%	—	85%	Promotion Effectiveness = DIVIDE(CALCULATE([Total Net Revenue...
Average Discount per Promo	RATIO	\$\$,##0	—	85%	Average Discount per Promo = DIVIDE(SUM('retail_fact_orders'...
On-Time Delivery Rate	RATIO	0.0%	✓	85%	On-Time Delivery Rate = DIVIDE(CALCULATE(COUNTROWS(retail_fa...
Average Days to Deliver	AVG	#,##0.0	✓	95%	Average Days to Deliver = CALCULATE(AVERAGE('retail_fact_ord...
Cancellation Rate	RATIO	0.0%	✓	90%	Cancellation Rate = DIVIDE(CALCULATE(COUNTROWS(retail_fact_o...
Shipping Cost as % of Revenue	RATIO	0.0%	—	100%	Shipping Cost as % of Revenue = DIVIDE(SUM('retail_fact_orde...
Average Customer Rating	AVG	0.0	✓	100%	Average Customer Rating = CALCULATE(AVERAGE('retail_fact_ord...
Revenue by Region	SUM	\$\$,##0	—	95%	Revenue by Region = CALCULATE(SUM('retail_fact_orders'[NetRe...
Customer Lifetime Value (CLV)	SUM	\$\$,##0	✓	90%	Customer Lifetime Value (CLV) = SUM('retail_fact_orders'[Net...
Month-over-Month Revenue Growth	CUSTOM	0.0%	✓	90%	Month-over-Month Revenue Growth = VAR CurrentMonthRevenue = ...
Units per Order	RATIO	#,##0.0	—	100%	Units per Order = DIVIDE(SUM('retail_fact_orders'[Quantity])...
Brand Revenue Share	RATIO	0.0%	—	90%	Brand Revenue Share = DIVIDE(SUM('retail_fact_orders'[NetRev...
Return Cost Impact	SUM	\$\$,##0	—	100%	Return Cost Impact = CALCULATE(SUM('retail_fact_orders'[NetR...
Payment Method Distribution	RATIO	0.0%	—	85%	Payment Method Distribution = DIVIDE(COUNTROWS(retail_fact_o...
Loyalty Tier Revenue	SUM	\$\$,##0	—	95%	Loyalty Tier Revenue = CALCULATE(SUM('retail_fact_orders'[Ne...

SECTION 05

Data Quality and Error Resolution

4 issue(s) were detected during file processing and mapping. 4 required user decisions; 0 were auto-resolved. All decisions are documented below and in the accompanying audit log JSON.

ID	Category	Severity	Table/Column	Resolution	Action	Status
62e772	Data Quality	HIGH	retail_fact_orders ShipDate	User Decision	Assign to "Unknown" placeholder	Resolved
003028	Data Quality	HIGH	retail_fact_orders DeliveryDate	User Decision	Assign to "Unknown" placeholder	Resolved
2db80c	Data Quality	HIGH	retail_fact_orders ReturnReasonCode	User Decision	Assign to "Unknown" placeholder	Resolved
07f17f	Data Quality	HIGH	retail_dim_returnreas ons ReasonCode	User Decision	Assign to "Unknown" placeholder	Resolved

User decision dialogue summaries

Null Join Key — retail_fact_orders.ShipDate: 6 rows (6.0%) in retail_fact_orders.ShipDate have null values. These rows cannot be joined to any dimension table. **User decision:** Assign to "Unknown" placeholder. 6 null values remapped to 'UNKNOWN' placeholder.

Null Join Key — retail_fact_orders.DeliveryDate: 8 rows (8.0%) in retail_fact_orders.DeliveryDate have null values. These rows cannot be joined to any dimension table. **User decision:** Assign to "Unknown" placeholder. 8 null values remapped to 'UNKNOWN' placeholder.

Null Join Key — retail_fact_orders.ReturnReasonCode: 94 rows (94.0%) in retail_fact_orders.ReturnReasonCode have null values. These rows cannot be joined to any dimension table. **User decision:** Assign to "Unknown" placeholder. 94 null values remapped to 'UNKNOWN' placeholder.

Null Join Key — retail_dim_returnreasons.ReasonCode: 1 rows (11.1%) in retail_dim_returnreasons.ReasonCode have null values. These rows cannot be joined to any dimension table. **User decision:** Assign to "Unknown" placeholder. 1 null values remapped to 'UNKNOWN' placeholder.

Audit trail	A complete machine-readable resolution log (error_resolution_log.json) is included with the deliverables. It records every error, all options presented, the user's choice, rows affected, and timestamps — suitable for GDPR lineage documentation and SOX audit requirements.
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SECTION 06

Semantic Model Summary

Component	Value
Tables in model	7
Relationships	9

Measures	30
Total rows	100
Build warnings	0
Built at	2026-07-08 18:02:25

Relationship graph

From	Column	To	Column	Type	Active
retail_fact_orders	DateKey	retail_dim_date	DateKey	MANY_TO_ONE	Yes
retail_fact_orders	CustomerID	retail_dim_customers	CustomerID	MANY_TO_ONE	Yes
retail_fact_orders	ProductID	retail_dim_products	ProductID	MANY_TO_ONE	Yes
retail_fact_orders	ChannelID	retail_dim_channels	ChannelID	MANY_TO_ONE	Yes
retail_fact_orders	PromoID	retail_dim_promotions	PromoID	MANY_TO_ONE	Yes
retail_fact_orders	ReturnReasonCode	retail_dim_returnreasons	ReasonCode	MANY_TO_ONE	Yes
retail_fact_orders	OrderDate	retail_dim_date	Date	MANY_TO_ONE	No
retail_fact_orders	ShipDate	retail_dim_date	Date	MANY_TO_ONE	No
retail_fact_orders	DeliveryDate	retail_dim_date	Date	MANY_TO_ONE	No

SECTION 07

Sign-off and Acknowledgement

This report is the formal completion record for Phase 1 of the **Finance** automation project. It documents all inputs received, processing decisions made, data quality resolutions agreed, and outputs generated.

Field	Value
Project name	Finance

Project ID	df0a6244-1430-483c-a3bd-4c9bed38b2e2
Pipeline version	AI Reporting Automation System v1.0
Phase completed	Phase 1 — Semantic Model Build
Session started	2026-07-08 17:57:21
Report generated	08 July 2026, 18:05
User decisions	4
Auto-resolutions	0
Pending issues	0
Phase 2 status	Available on request

This report was automatically generated by the AI Reporting Automation System v1.0 upon completion of the semantic model build pipeline. All data quality decisions were made interactively through the chat interface by an authorised user and are recorded in the accompanying audit log.
Report ID: DF0A6244-CR1 • 08 July 2026, 18:05 • CONFIDENTIAL